



Helping Users Understand Their Spending — Not Control It

A behavioral exploration inside Google Pay

Understanding the Landscape: Decoding Personal Finance

Google Pay's Global Surge and Market Dominance

User Stats: Google Pay boasts 820 million active users globally in 2025, with 165 million in the US and India comprising 62% of the base. 85% of users are aged 18–45, with millennials at 45% and Gen Z adoption rising 30%.

Scale: Handles over 2.1 billion transactions worth \$1.32 trillion annually, a 30% YoY increase in volume; average transaction is \$76, with P2P payments exceeding \$370 billion.

Market Share: Captures 18% of the global digital payments market and 21.5% of mobile wallets, trailing Apple Pay but ahead of Samsung Pay; 20% of all mobile transactions worldwide.

Footprint and Context: Operates in 45+ countries supporting 60+ currencies, with 6.4 million integrated merchants; strong growth in emerging markets like India (\$610B in transactions) and Southeast Asia (46% YoY), emphasizing its Android dominance (65% of users).

Income Sources of Gpay

Primary Sources: Collects fees from recharges, bills, and bank/merchant transactions (\$4.1B globally, ₹14.8 crore India FY21), monetizes via in-app ads and sponsored promotions (\$0.4B globally), splits earnings from bank, e-commerce, and payment partnerships (e.g., virtual cards).

Market Edge: Leverages 150M users worldwide (67M India; 35% UPI share) for strong network effects, integrates with Google tools (Chrome, Assistant, Maps) and adapts locally (UPI, languages, rural programs), enhances loyalty through security (encryption, biometrics) and perks (rewards, cashbacks).

Problem Statement

Google Pay reveals spending totals, but hides the 'why'—fueling confusion, guilt, and app abandonment. **Most finance apps show numbers, not meaning**

Market Landscape: How Finance Apps Explain Spending (Research Findings)

What apps show today

Over-reliance on Charts
Generic Alerts

("You spent 80% of budget")

Backward-Looking Data (reports on past mistakes)

What's missing

Lack of Context
(ignores life events or emergencies)
No Behavioral Patterns (ignores "why" you spend)
Emotional Logic (fails to address stress-spending)

User outcome

Information Overload
causing mental fatigue
Financial Guilt and shame-based avoidance
Low Engagement (app opened only in "crises")

The Data-Meaning Gap: Insights from 35 Real Users

GPay Research Spotlight "I see the numbers, but I don't see the **why**." Our survey of 35 users confirms: **68%** still struggle to understand why they overspent, proving that current insights lack behavioral context.

CLICK FOR MORE INFO

Segmentation and Hypothesis

UX Journey Audit

• Transaction

Seamless spending with UPI/QR

• Insight

Transaction history, ID
Merchant codes,
Balance check—
"Where did all the salary go, its just 15th of the month?"

• User Reaction

Guilt, Anxiety,
Fear...zero
spending context

Total
avoidance

Disengagement

Unmet Need: Translating transactional history into a behavioral narrative that the human brain actually recalls, planning finances better.

Focus Segment: "Frictionless Fog" Spenders

High-frequency UPI users who have mastered *how* to pay but lost *why* they spend. They rely on GPay for everything—from ₹20 chai to ₹2000 shopping—creating a mental disconnect between tapping and value loss. **SURVEY (N=35):**

54% (n=19) form the core segment, **100%** of this group use GPay daily or multiple times a day

69% of total users (nearly all in this segment) question where money went at month-end

56% report insight fatigue: GPay shows numbers, not reasons—charts fail to drive understanding or behavior change

Impact if Solved

Trust & Sentiment: Shifts users from *shame-based avoidance* to *empowered reflection* by explaining **why** spends happen (stress, routine, necessity).

Retention & LTV: Directly addresses the **31.4% insight-fatigued users**, keeping them engaged with Insights instead of abandoning review.

Ecosystem Depth: Users who feel *understood* (not judged) are more likely to adopt **credit, insurance, and investments**, strengthening long-term platform value.

Why this Segment?

Active Majority: Highest transaction volume, highest post-spend anxiety.

Utility > Awareness: Fast payments, weak reflection ("utility-high, awareness-low").

Tool Mismatch: Existing charts help budgeters, not these users.

Review Avoidance: ~28% "ignore and move on" after overspending → broken feedback loop.

Business Impact: Solving this prevents disengagement and positions GPay as supportive, not just a transactional tool.

Core Hypothesis: Failing Spending Insights

H1: Context Gap

Users remember *situations*, not categories. GPay stores spends as merchants/tags, but human recall is event-based ("after work stress", "mid-month travel"). This mismatch breaks understanding.

H2: Narrative Absence

Without behavioral framing, users must mentally reconstruct patterns, increasing cognitive load and avoidance.

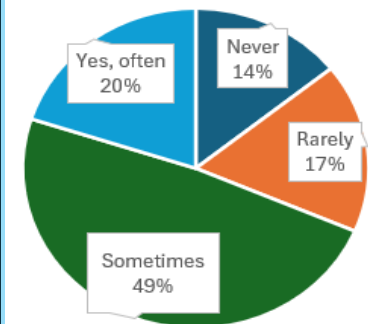
H3: Recall Breakdown

When users cannot explain a spend without effort, reflection fails. Over time, abandonment of App.

User Research Insights and Persona Building

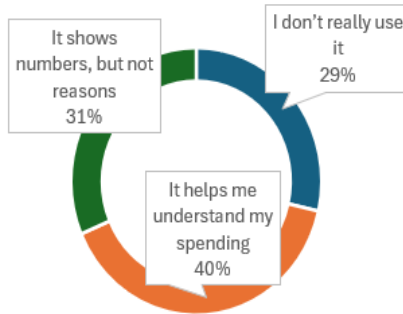
User Research Insights: Surveys, Interviews & Recall Walkthroughs

Awareness Gap



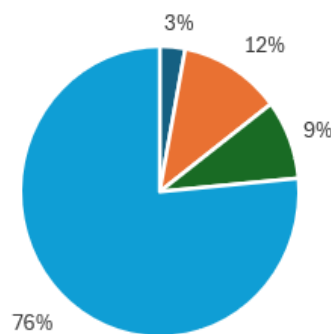
~69% experience "Transaction Amnesia," frequently losing track of the specific reasons/ life events behind their spendings.

Data vs. Meaning



~31% find current data technically accurate but humanly empty, reports that the app "shows numbers, but not reasons"...

Spending Analysis



After a month of overspend, 76% try to analyze but fail, 12% ignore, 9% set rules, 3% disengage emotionally.

Core Interview Insights

- **The Identity Crisis:** Abstract registered names (e.g., "Avion Enterprise") break the memory link to life events (e.g., "Mother's Birthday Air Fryer"), causing user embarrassment and frustration.
- **Sales vs. Sense-Making:** Users feel the "Manage Money" section is a mismatch; it prioritizes selling loans over categorizing Investments, Insurances, EMIs...
- **The Cumulative Blindspot:** Small, frequent transactions (₹10–20) are invisible individually but collectively drain "miscellaneous" budgets without visual grouping.
- **Defensive Avoidance:** High cognitive effort in decoding cryptic transaction lists leads users to accept ignorance rather than review history.



Anita Bannerjee – 24 - Student

Anita (24) is a digitally native, frequent Google Pay user from the *Transactional Reflexive* segment. Despite high usage, she experiences **transaction amnesia**. In a recall walkthrough, she couldn't remember a ₹7,500 payment from seven months ago because GPay showed the legal entity name "**Avion Enterprise Group**" instead of the familiar store "**Taj Electronics,**" breaking her memory link to the gift purchase.

JTBD:

- Instantly recognize past spends using remembered names/events to retain financial agency
- Reduce cognitive load while reviewing history, avoiding manual investigation or family calls

Friction: Merchant-name confusion triggers surprise, embarrassment, and mild anxiety.

Insight: Users remember **events and stories**, not legal merchant entities.



Poushali Dutta - 33 – Data Scientist (Working Mom)

Poushali (33) data scientist uses Google Pay daily for a range of transactions—food delivery, OTT subscriptions, utility bills, and ad-hoc payments. Highly tech-savvy, expects her finance apps to reflect the same logical structure and clarity she applies in her work. Instead, she experiences GPay as fragmented: rich in raw data, but poor at helping her make sense of it.

Primary Jobs-to-be-Done (JTBD)

Intent-based categorization: Help her see spending grouped by purpose (e.g., food delivery, travel), not just merchant names.

Behavioral reflection: Enable tracking *Investments, Insurances, Loans...*

Psychological Barriers: Finds the "Manage Your Money" section misaligned—it pushes loans instead of insights. Payments under "Businesses," like one random OLA ride under name "Omprakash Shah," lack context, leaving her unsure why the spend happened or how it should be interpreted.

Core Behavioral Insight: Google Pay executes transactions well, but fails to explain them meaningfully.



The Problem Framing Canvas: Breaking Down the True Problem

What is the True Problem?

High-frequency GPay users lose financial awareness not because data is missing, but because context is. GPay captures every transaction accurately, yet presents spending as raw merchant names and amounts, without the situational context that users remember, turning routine review into cognitive labor. Dependence on multiple platforms for financial overview.

Small, frequent transactions remain invisible individually but accumulate into mid-month stress and confusion. Users feel overwhelmed, judged by numbers they can't explain, and begin avoiding financial review altogether. The problem is not lack of control—it is lack of clarity...

Who is facing it?

Problem primarily faced by high-frequency users who rely on Gpay for everyday payments. Digitally comfortable but cognitively overloaded, often depending solely on GPay to recall spends; irregular income earners, family finance overseers...Solving for missing context and meaning—not data, controls, or user discipline.

How do we know this is a real problem?

This is a real problem because users actively create workarounds to compensate for missing clarity. ~69% experience transaction amnesia, and ~31% say insights are technically correct but meaningless.

Interviews and recall walkthroughs showed frequent confusion over merchant names, forgotten high-value spends, and manual reconstruction via other apps or people. ~28% ignore or swipe away insights to avoid guilt, signaling disengagement.

When users alter behavior to bypass a feature, it indicates a systemic UX failure—not a minor usability edge case.

Value generated by solving this problem

A. For Users

Lower Cognitive Load: No need to remember who sent what or when
Clarity over Control: GPay already equipped with robust security features shifts to a trusted *financial memory*, explains *why* money moved clear and guilt-free

B. For Business (Why GPay is uniquely positioned)

First-Party Advantage: Owns real transaction data; no app-switching required
Session Depth: Users stay in GPay instead of Drive/Notes/other finance apps
Commerce Enablement: Reliable retrieval builds trust for Pay, credit, insurance, investments...
Differentiation: Moves beyond cluttered hubs or rigid budgeting tools into *sense-making, user-friendly finance buddy*.
Scales Naturally: Works even for irregular income earners, young professionals without forcing discipline...

The Urgency to Solve

Competitors in India like Paytm, CRED, and PhonePe are rapidly layering AI-driven money tools. Paytm now auto-tags and summarizes spending, while CRED Money aggregates accounts to surface actionable insights...

Google Pay, despite its massive 530M+ Indian user base, is still primarily transactional. While it has introduced voice payments and UPI-linked credit cards with basic tracking, it lags behind competitors in contextual financial understanding...

With GPay's UPI share dropping from ~46% to ~34%, the urgency is clear: without meaningful insights, GPay risks user churn and loses monetization opportunities from higher-value financial products...

Ideation – Solution Framing

Solution A: Proactive Spend Guard

(Behavior Monitoring Layer)

Idea

A real-time system that observes spending behavior and intervenes during transactions to discourage “unnecessary” purchases.

Key Features

Live prompts during payment moments (“You’ve ordered food 3 times this week”)
Category-based monthly thresholds (food, shopping, entertainment)
Alerts on deviation from historical averages
Generic finance nudges (save more, delay purchase, invest instead)

Why this Solution

Explores whether in-the-moment awareness can influence spending decisions by surfacing behavioral patterns at the point of action.

Solution B: Smart Finance Manager

(Advice-First Financial Assistant)

Idea

A centralized AI financial manager that aggregates income, expenses, investments, loans, and goals to guide users toward “better” financial outcomes.

Key Features

Auto-categorization: food, transport, subscriptions, utilities, EMIs
Investment suggestions (SIP, FD, NPS, tax-saving)
Credit card insights (limits, rewards, benefits)
Goal tracking (retirement, emergency fund, insurance)

Why this Solution

Addresses fragmentation by consolidating financial data and offering structured guidance across spending, saving, and investing.



Solution C: The Spending Translator

(Reflective Intelligence Layer + Interface Innovation)

Idea

A calm, opt-in intelligence layer that translates raw transactions into human-understandable context—helping understand *context*, without telling them what to do.

Key Features

Contextual Categories

Replaces “People / Businesses” with: Food Delivery, Transport, Subscriptions, Utilities, EMIs & Credit, Fuel, ATM, Entertainment

Pattern-First Visuals

Line charts (daily / weekly / monthly). Neutral annotations for spikes, for clarity. No red flags, no totals-first framing

AI Chat Bot

Answers natural questions like:

“Why did July feel more expensive than June?” “Do I spend more on food when traveling?”

“How did small ₹10–₹20 spends add up?”

“last date for taxes” “benefits of SBI credit card”

Financial Context (Not Advice)

Single view of assets: bank balance, investments, gold, loans, EMIs

Calculators (SIP, FD, NPS, EMI) via dropdown

Credit cards listed with features & limits

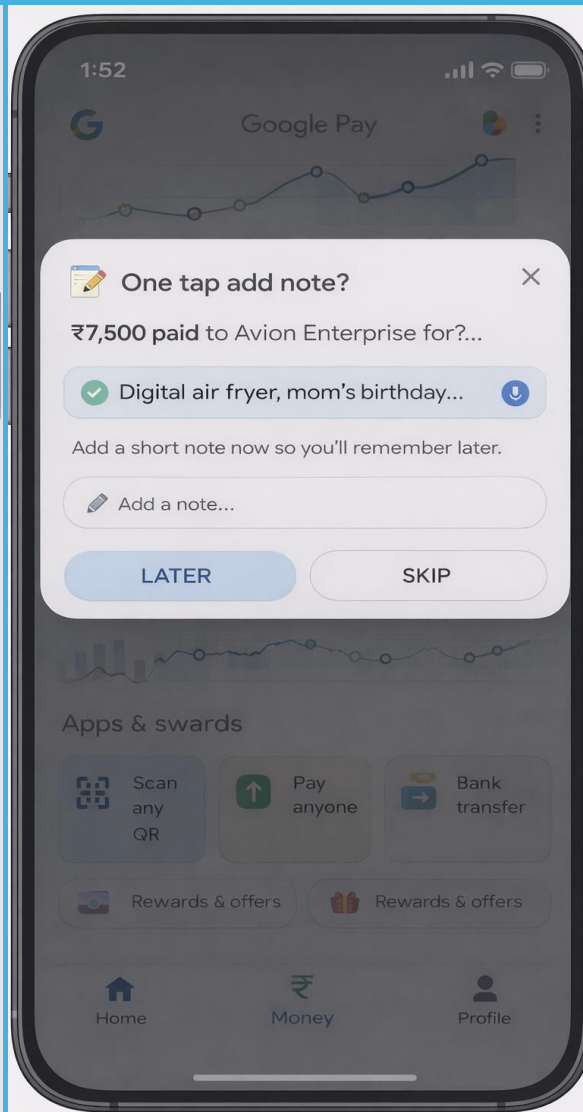
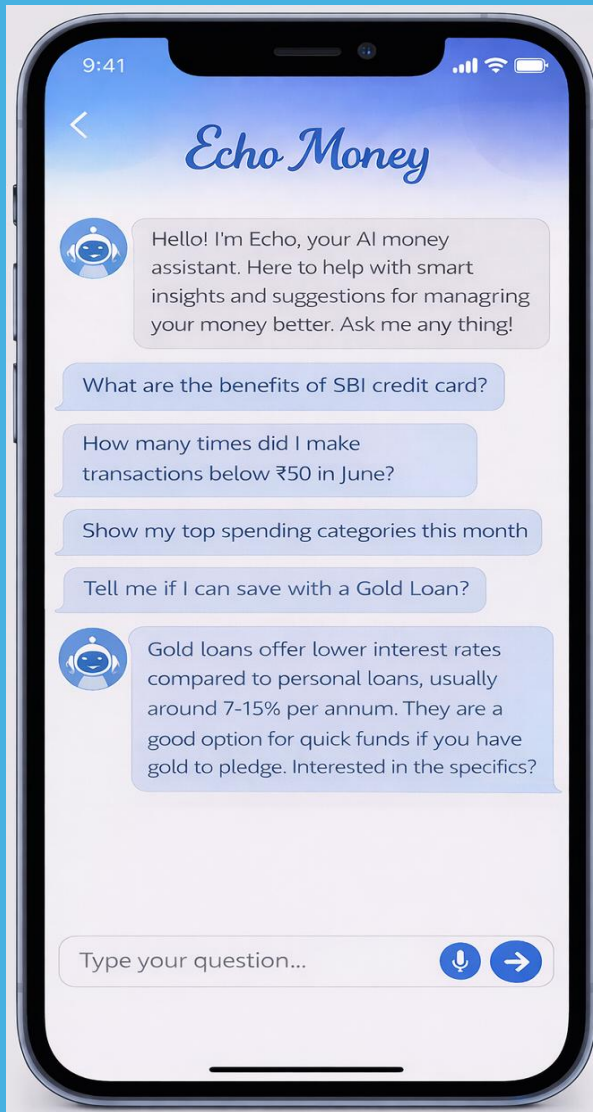
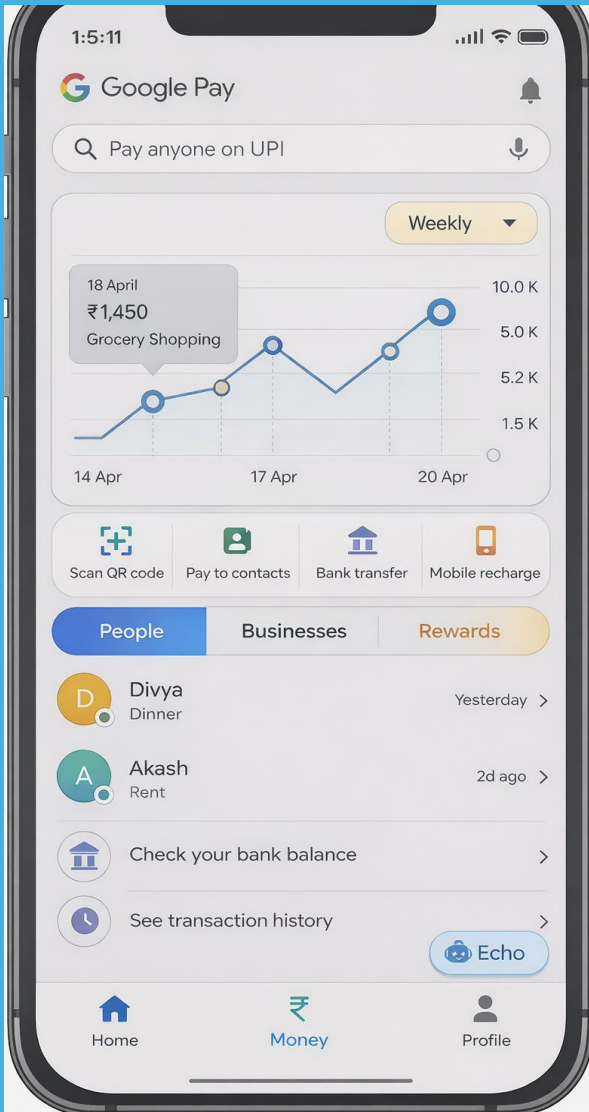
Why this Solution

Designed to support reflection and self-awareness by translating financial data into narratives users recognize—without judgment, control, or prescription.

Prioritization Framework

Criteria	Solution A: Proactive Spend Guard	Solution B: Smart Finance Manager	★ Solution C: The Spending Translator
Core Problem Addressed	Perceived Overspending. Helps users stop spending via friction.	Cognitive Load . Consolidates multiple financial data points.	Lack of Awareness and Clarity. Makes raw transaction data recognizable by building contextual categories and support provided by AI Chat Bot.
Solving User Need	Low. Triggers "Guilt" and avoidance behaviors.	High. Useful financial categorization, structured guidance and insights.	High. Directly solves "Amnesia" & "Logic Gaps" without judgment. Total financial clarity and self awareness at user's discretion.
Activation	Low. It is a restrictive feature that may lead to churn.	Medium. High utility but lacks an emotional "Hook."	Very High. Creates a "Financial Memory Bank," increasing stickiness. Voluntary, intent-driven engagement.
Feasibility	High. Standard rule-based "If-Then" logic.	Medium. Requires heavy Account Aggregator (AA) integration.	Medium. Requires on-device AI (SLM) for contextual tagging, neural annotations for line charts.
Risk	High. Users may feel over-controlled and may uninstall.	Medium. Privacy concerns regarding data aggregation using AI. AI may also hallucinate, risk of incorrect inference.	Medium. Potential misinterpretation of context if patterns are inferred without full user input. Privacy perception risk if users feel their transactions are being “analyzed” without clarity. Requires strong transparency, on-device processing, to maintain trust
Rejected/ Selected	Rejected — Violates emotional safety; induces guilt and avoidance, might feel like monitoring.	Rejected — Shifts from reflection to prescription; more of a financial guidance tool; overlaps with existing finance apps.	Selected — Maximizes clarity, trust, and self-agency without behavioral control.

Wireframe & Flow : The User Journey (Part 1: The Home Tab)



Screen 1: Home (Reflective Overview)

User action: User opens GPay and sees a spending line graph with a time filter (daily/weekly/monthly/yearly). Tapping a spike reveals related spends. Echo ChatBot is always visible.

Backend: During idle states (charging/Wi-Fi), transactions are indexed on-device. Category tags and user notes are stored in a local database. Aggregated patterns drive graph markers with near-zero latency.

Screen 2: Echo Money (AI Reflection)

User action: User clicks on the Echo button on Home screen, asks natural questions about past spends or patterns.

Backend: Echo queries locally stored summaries using text + semantic matching, not raw logs.

Screen 3: Gentle Reminder (Post-Transactions)

User action: After payments sent or received—especially unknown merchants—a soft “Add note?” card appears (type, voice, skip, defer).

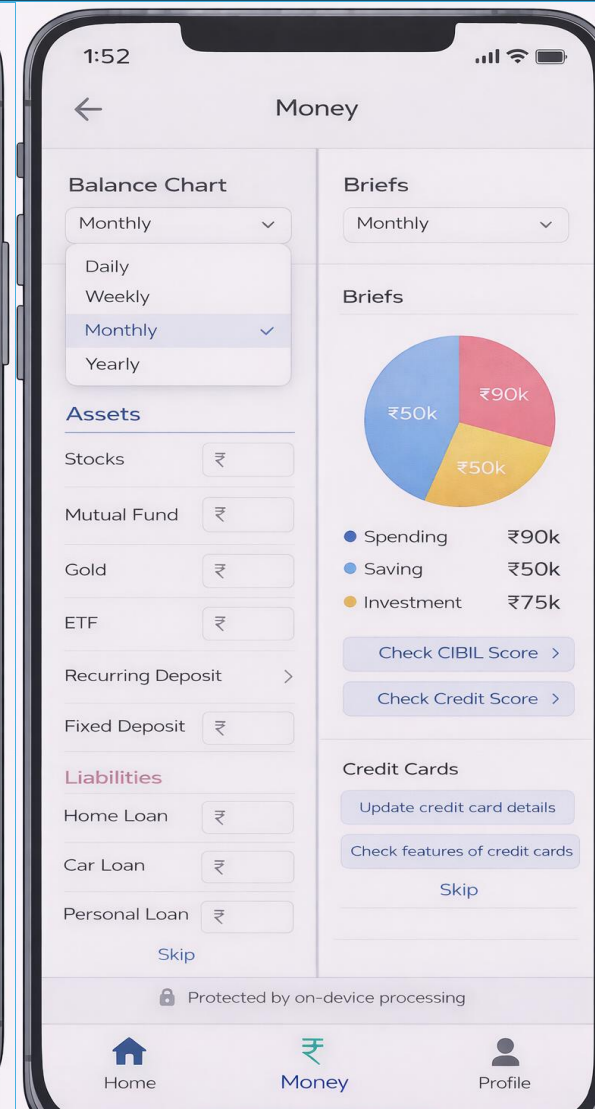
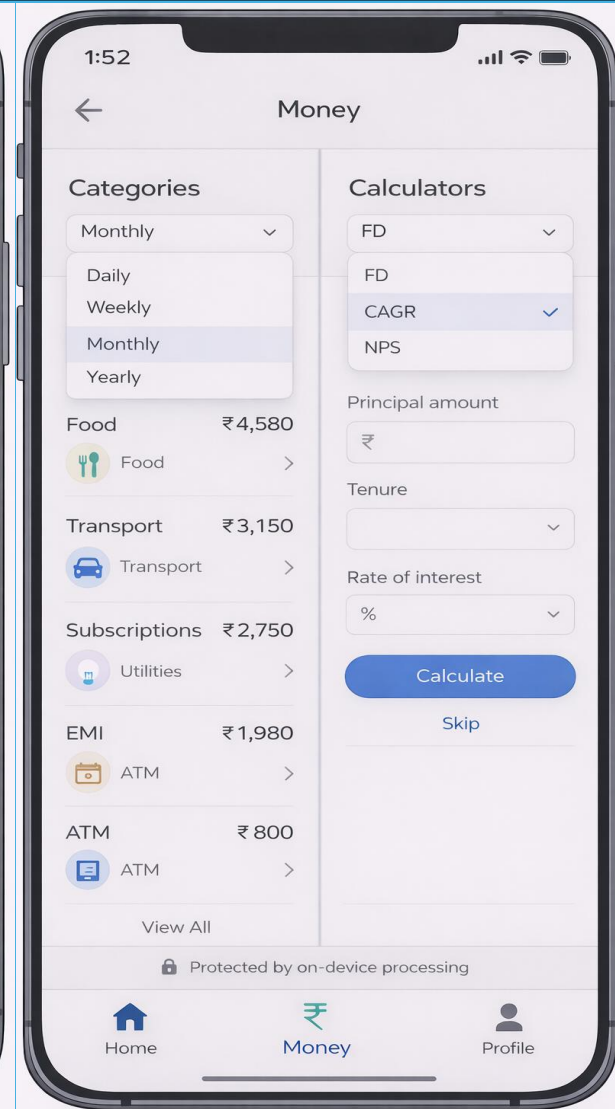
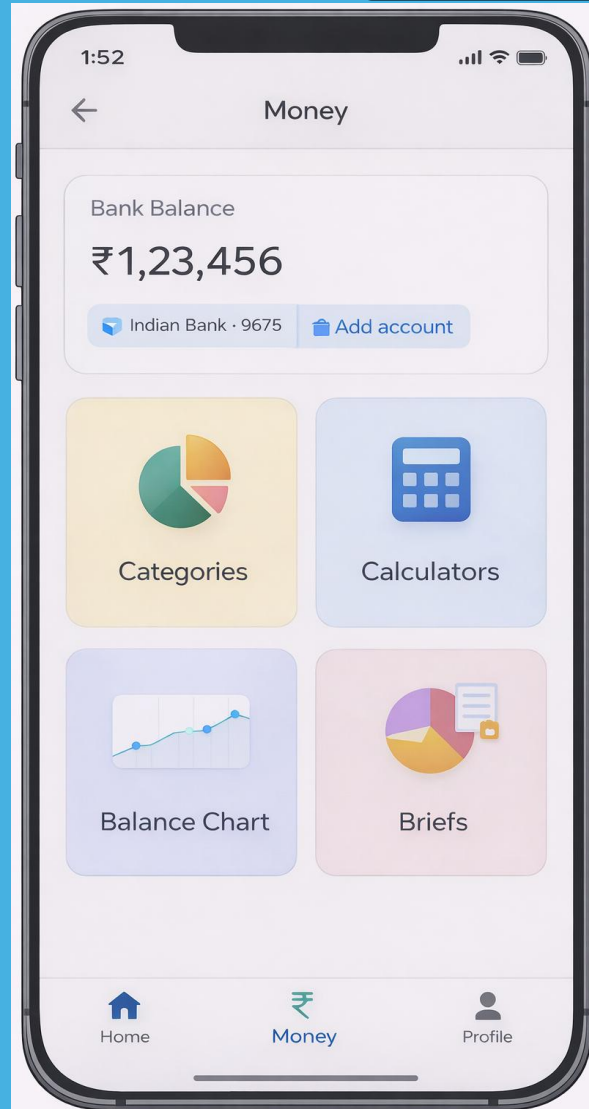
Backend: Notes attach locally to transactions, improving future search, categorization, and annotations.

Screen1:Home /Reflective Overview
Line chart with **context markers** on key spends...
Echo button, AI ChatBot always visible, not intrusive

Screen 2 : Echo Money/
Chat Style AI Reflection Space
Ask any question...
Get **clear, neutral answers** on any financial query

Screen 3 – Gentle Reminder/
Adding Context after transactions
Optional **one tap add note / voice note...**
One-tap **Skip or Later**, no friction

Wireframe & Flow : The User Journey (Part 2: The Money Tab)



Screen 4: Money Tab

User action: User enters the Money tab for a calm financial overview. Taps any of the four cards to explore details—no prompts or warnings.

Backend: Pre-aggregated summaries are fetched from the local database for instant, low-latency load.

Screen 5: Categories & Calculators

User action: User taps a category to view spends by day/week/month. Calculators are used only when needed and can be skipped.

Backend: Transactions are AI-tagged locally using merchant data and user notes. Queries run on precomputed clusters, fully on-device.

Screen 6: Balance Chart & Briefs

User action: User optionally adds assets and liabilities. Briefs show a pie split of spending, saving, and investment; credit & CIBIL score; credit card details & features can be viewed or skipped.

Backend: All inputs remain on-device. Briefs are generated from local aggregates—no advice, no scoring, just clarity.

Screen 4 – Money Tab

Financial snapshot, all insights under one roof...
Four cards: Categories, Calculators, Balance Chart, Briefs.

Screen 5 – Categories + Calculators

Categories show spend by type. Tap any category to deep dive by time range drop-down...Calculators are optional, skippable, self-initiated.

Screen 6 – Balance Chart + Briefs

Assets/Liabilities view (edit/ skip) Briefs give a quick pie view of Spend/ Save/Invest by time range drop-down. Credit, CIBIL score, Credit card.

Guardrails: Measuring Success & Ensuring Trust

Success Metrics

North Star Metric: Recall Confidence Rate

Definition: % of users who can correctly identify transaction intent/context within 3 seconds.

Target: Increase from **31% (Baseline)** to **70–75%** within 2 months.

Leading Indicators (Early Signals)

Proactive Interaction: % of DAU voluntarily opening "Translations" or asking the AI assistant for context (Target: 25–30%).

Label Correction Rate: % of users manually editing AI-generated categories (Target: <5% to ensure high semantic accuracy).

Lagging Indicators (Long-term Impact)

Avoidance Reduction: % drop in users dismissing insights immediately due to guilt or discomfort (Baseline: **28%** → Target: **<15%**).

Search Efficiency: Measurable reduction in self-reported "Time-to-Find" specific past transactions compared to legacy keyword search.

Guardrail Metrics (Trust & Performance)

Context Latency: Load time for explanations stays **<150ms** (Ensuring it feels native and instant, AI processing does not slow down UI).

Sentiment Score: Qualitative ratio of feedback using terms like **“clear,” “helpful,”** and **“non-judgmental”** vs. “creepy” or “intrusive.”

Pitfalls & Mitigations

Risk 1: Perceived Surveillance

Users may feel their spending is being closely watched or evaluated.

Mitigation: On-device processing by default. All interpretation happens locally. Clear cues like “Processed on your phone” signal no central analysis, preserving trust and control.

Risk 2: Misinterpreted Context

The system may infer the wrong intent or category.

Mitigation: Confidence-aware interpretation. Low-confidence matches use soft language (“This may be...”) with a one-tap correction to quickly refine context.

Risk 3: Cognitive Overload

Excess explanation may overwhelm rather than help.

Mitigation: Progressive disclosure. Insights stay hidden by default and appear only when users tap or ask, keeping the main view clean.

Risk 4: Loss of Emotional Safety

Users may feel judged or subtly pushed to “improve.”

Mitigation: Non-prescriptive behavior. The system describes patterns without advice, warnings, or corrective language—supporting understanding, not control.

Product Ethics: Reflective insights are explicitly opt-in; the core payment experience remains unchanged. The system is strictly non-prescriptive—no advice, scores, warnings, or nudges—only neutral pattern explanations. All context is processed fully on-device, with no inferred narratives stored or shared externally. Users retain full control and can ignore, edit, or dismiss insights at any time, preserving emotional safety and trust.

Research Ethics: All interviews and recall walkthroughs were conducted with informed consent, clearly stating that spending behavior would be discussed. No real financial data (screenshots, balances, account details) was collected or stored. All names, merchants, and identifiers were anonymized; personas are fictionalized from aggregated patterns. Research insights were used solely to understand behavioral gaps, not to profile or evaluate individuals.